

LORAIN COUNTY REGIONAL, OH, USA

WMO: 725214

Lat: 41.346N

Lon: 82.179W

Elev: 242

StdP: 98.45

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 04849

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.7	-13.5	-21.2	0.6	-16.0	-18.1	0.8	-12.7	12.6	0.7	11.8	-1.1	3.8	250	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	32.1	23.0	30.8	22.4	29.1	21.6	24.6	29.7	23.7	28.4	22.8	27.5	4.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	18.1	27.6	22.3	17.5	26.6	21.3	16.4	25.7	75.4	29.5	71.6	28.5	68.6	27.5	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.2	9.7	8.6	-20.1	34.1	3.5	1.4	-22.6	35.0	-24.7	35.8	-26.6	36.6	-29.1	37.6
			-20.5	26.0	3.4	1.1	-22.9	26.8	-24.9	27.4	-26.8	28.1	-29.3	28.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	-2.8	-1.6	3.1	9.9	16.1	20.9	22.7	21.6	18.4	12.2	6.0	0.3
	DBStd	10.31	6.52	6.24	6.05	5.32	4.80	3.82	3.04	2.86	3.85	4.81	5.25	5.64
	HDD10.0	1505	399	328	228	67	6	0	0	1	31	142	303	
	HDD18.3	3251	657	559	472	259	103	19	2	6	46	199	369	559
	CDD10.0	1732	1	3	15	63	194	326	393	360	252	99	23	3
	CDD18.3	434	0	0	1	5	33	96	137	107	48	8	0	0
	CDH23.3	3663	0	0	4	54	322	788	1207	823	397	66	2	0
(14) Wind	WSAvg	4.0	5.1	4.8	4.7	4.7	3.9	3.5	2.9	2.7	2.9	3.8	4.5	4.8
	PrecAvg	942	59	51	68	89	100	96	99	87	86	62	78	69
	PrecMax	1226	155	121	151	228	263	201	217	202	234	145	188	204
	PrecMin	491	9	5	16	15	15	22	27	14	15	12	18	27
	PrecStd	143	33	29	28	39	52	38	42	45	47	28	39	30
	DB	15.9	18.8	22.7	27.8	31.3	33.0	34.0	32.5	32.4	27.9	22.1	17.0	
	MCWB	12.8	13.6	15.5	18.0	21.2	22.5	24.4	23.8	22.6	20.2	14.9	13.2	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	12.5	13.7	18.8	24.8	29.1	31.5	32.2	30.8	29.6	25.2	18.5	13.7	
	2% MCWB	10.3	10.3	13.2	16.0	19.9	22.3	23.5	23.0	21.7	18.2	14.1	10.8	
	5% DB	9.0	10.3	16.1	22.2	27.3	29.7	30.7	29.0	27.5	22.6	16.5	11.0	
	5% MCWB	6.8	7.4	11.9	15.1	19.3	21.4	22.7	22.3	20.5	16.7	12.2	8.5	
	10% DB	6.2	7.0	12.9	19.1	24.7	27.7	28.9	27.6	25.6	20.2	14.1	8.0	
	10% MCWB	3.9	4.2	9.2	13.2	17.9	20.7	21.9	21.5	19.5	15.5	10.5	6.1	
	WB	13.3	14.0	16.6	19.5	22.9	24.6	26.1	25.5	24.0	21.5	17.2	13.9	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	15.4	17.9	21.7	25.2	29.1	30.3	32.4	30.5	29.7	27.1	20.1	16.3	
	2% WB	10.5	10.8	14.3	17.5	21.3	23.5	25.0	24.2	22.8	19.3	14.8	11.4	
	2% MCDB	12.3	13.0	17.9	23.1	27.2	29.1	30.5	28.5	27.7	23.4	17.9	13.4	
	5% WB	7.1	7.4	12.2	15.7	20.2	22.5	23.8	23.3	21.6	17.7	12.9	8.5	
	5% MCDB	8.9	9.9	15.8	21.0	25.4	28.0	28.5	27.5	25.9	21.3	15.8	10.2	
	10% WB	3.9	4.1	9.5	13.9	18.9	21.6	22.9	22.4	20.5	16.1	11.0	6.3	
	10% MCDB	6.0	6.7	12.7	18.5	23.5	26.5	27.3	26.3	24.3	19.5	13.8	8.2	
(35) Mean Daily Temperature Range	MDBR	7.4	8.0	8.7	10.5	10.9	10.1	10.3	10.3	11.0	9.8	8.3	6.6	
	5% DB	10.2	12.1	13.4	14.6	13.6	12.5	11.8	11.9	13.3	13.3	11.7	9.6	
	5% MCWBR	8.9	9.8	9.2	8.7	6.9	5.7	5.3	6.0	6.8	7.6	8.3	8.1	
	5% WB	9.9	11.5	12.4	13.2	11.8	10.7	10.0	9.9	10.9	11.5	10.3	9.4	
	5% MCWBR	9.0	9.9	8.9	8.9	6.8	5.7	5.2	5.5	6.1	7.1	8.2	8.7	
(40) Clear-Sky Solar Irradiance	taub	0.305	0.334	0.356	0.393	0.428	0.445	0.443	0.429	0.383	0.365	0.328	0.311	
	taud	2.406	2.289	2.312	2.276	2.215	2.213	2.241	2.266	2.371	2.399	2.466	2.451	
	Ebn at Noon	847	867	890	878	851	835	832	833	851	823	813	808	
	Edh at Noon	80	105	115	129	140	141	136	129	108	94	76	72	
(44) All-Sky Solar Radiation	RadAvg	1.40	2.24	3.35	4.54	5.40	5.98	6.01	5.32	4.27	2.77	1.75	1.17	
	RadStd	0.13	0.30	0.31	0.37	0.46	0.42	0.36	0.23	0.38	0.23	0.27	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-0.94	N/A	N/A	N/A	N/A
(47)	Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+108	N/A

Nomenclature: See separate page